

Complete flatmounting of the macaque cerebral cortex.

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The elaborate folding of the brain surface has posed a practical impediment to investigators engaged in mapping the areas of the cerebral cortex. This obstacle has been overcome partially by the development of methods to erase the sulci and gyri by physically flattening the cortex prior to sectioning. In this study, we have prepared a step-by-step atlas of the flatmounting process for the entire cerebral cortex in the macaque monkey. The cortex was dissected from the white matter, unfolded, and flattened in a single piece of tissue by making three relieving cuts. The flatmount was sectioned at 60-75 microm and processed for cytochrome oxidase (CO) or myelin. From animal to animal there was nearly a twofold variation in the surface area of individual cortical regions, and of the whole cortex. In each specimen, a close correlation was found between V1 surface area (mean = 1343 mm²), V2 surface area (mean = 1012 mm²), hippocampal area (mean = 181 mm²), and total cerebral cortex area (mean = 10,430 mm²). The complete pattern of CO stripes in area V2 was labeled clearly in several cases; the number of cycles of thick-pale-thin-pale stripes ranged from 26 to 34. Characteristic patterns of strong CO activity were encountered in areas V3, MT, auditory and somatosensory cortex. In some animals we made injections of a retrograde tracer, gold-conjugated cholera toxin B subunit, into area V2 to identify all sources of cortical input. In addition to previously described inputs, we identified three new regions in the occipitotemporal region that project to V2. Flatmounting the cerebral cortex is a simple, efficient method that can be used routinely for mapping areas and connections in the macaque brain, the most widely used primate model of the human brain.

Cortical laminar necrosis following myocardial infarction.

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The cortical laminar necrosis (CLN) is a permanent injury characterized by the selective delayed necrosis of the cerebral cortex, mainly of the third layer, and usually greater in the depths and sides of the sulci than over the crest of the gyri. The damage involves all cellular components - either neurons, glia cells and blood vessels - and results in a focal cortical band of pan-necrosis detectable in late sub-acute or chronic stages of reduced energy supply to the brain. The CLN has been described in different conditions as hypoxia, hypoglycemia and status epilepticus. At brain CT or MR scans it appears with pathognomonic highly hyperdense or T1-hyperintense lesions following the gyral anatomy of the cerebral cortex. We reported a case of CLN associated to myocardial infarct and discussed the underlying mechanisms.

Quantitative differences in beta/A4 protein subtypes in the parahippocampal gyrus and frontal cortex in Alzheimer's disease.

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The density of diffuse, primitive and classic beta/A4 protein deposits was estimated in sulci and gyri in the frontal cortex and parahippocampal gyrus (PHG) in 8 cases of Alzheimer's disease. Total beta/A4 deposit density was similar in the frontal cortex and PHG but the ratio of primitive and classic deposits to the total was greater in the PHG compared with the frontal cortex. Total beta/A4 deposit density was greater in the depths of the sulci, but the proportions of the various beta/A4 subtypes were similar in

sulci and gyri. Hence, increased density of primitive and classic deposits in the PHG could reflect enhanced conversion of diffuse to mature deposits whereas increased density of mature beta/A4 subtypes in sulci versus gyri may reflect increased beta/A4 deposition in the sulci.

The Mid-Fusiform Sulcus (sulcus sagittalis gyri fusiformis).

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In the human brain, the mid-fusiform sulcus (MFS; sulcus sagittalis gyri fusiformis) divides the fusiform gyrus (FG) into lateral and medial partitions. Recent studies show that the MFS is identifiable in every hemisphere and is a landmark that identifies (a) cytoarchitectonic transitions among four areas of the FG, (b) functional transitions in many large-scale maps, and (c) the location of fine-scale functional regions. Thus, simply identifying the MFS in a person's brain provides researchers with knowledge regarding: (a) how cells are organized across layers within a particular cortical location, (b) how functional representations will be laid out in cortex, and (c) the precise location of functional regions from cortical folding alone. The predictive power of the MFS can guide future studies examining the anatomical-functional organization of the FG, as well as the development of translational applications for different patient populations. Nevertheless, progress has been slow in incorporating the MFS into the broader anatomical community and into neuroanatomical reference sources. For example, even though the MFS is a rare structural-functional landmark in human association cortex as just described, it is not recognized in the recently published Terminologia Neuroanatomica (TNA). In this review, I collate the anatomical and functional details of the MFS in one place for the first time. Together, this article serves as a comprehensive reference regarding the anatomical and functional details of the MFS, as well as provides a growing number of reasons to include the MFS as a recognized neuroanatomical structure in future revisions of the TNA. Anat Rec, 302:1491-1503, 2019. © 2018 American Association for Anatomy.

Temporal Variability of Cortical Gyrus-Sulcal Resting State Functional Activity Correlates With Fluid Intelligence.

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The human cerebral cortex is highly convoluted as convex gyri and concave sulci. In the past decades, extensive studies have consistently revealed substantial differences between gyri and sulci in terms of genetics, anatomy, morphology, axonal fiber connections, and function. Although interesting findings have been reported to date to elucidate the functional difference between gyri and sulci, the temporal variability of functional activity, which could explain individual differences in learning and higher-order cognitive functions, and as well as differences in gyri and sulci, remains to be explored. The present study explored the temporal variability of cortical gyrus-sulcal resting state functional activity and its association with fluid intelligence measures on the Human Connectome Project dataset. We found that the temporal variance of resting state fMRI BOLD signal was significantly larger in gyri than in sulci. We also found that the temporal variability of certain regions including middle frontal cortex, inferior parietal lobe and visual cortex was positively associated with fluid intelligence. Moreover, those regions were predominately located in gyri rather than in sulci. This study reports initial evidence for temporal variability difference of functional activity between gyri and sulci, and its association with fluid intelligence measures, and thus provides novel insights to understand the mechanism and functional

relevance of gyri and sulci.

A model of cerebral cortex formation during fetal development using reaction-diffusion-convection equations with Turing space parameters.

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The cerebral cortex is a gray lamina formed by bodies of neurons covering the cerebral hemispheres, varying in thickness from 1.25 mm in the occipital lobe to 4mm in the anterior lobe. The brain's surface is about 30 times greater than that of the skull because of its many folds; such folds form the gyri, sulci and fissures and mark out areas having specific functions, divided into five lobes. Convolution formation may vary between individuals and is an important feature of brain formation; such patterns can be mathematically represented as Turing patterns. This article describes how a phenomenological model was developed by describing the formation pattern for the gyri occurring in the cerebral cortex by reaction diffusion equations with Turing space parameters. Numerical examples for simplified geometries of a brain were solved to study pattern formation. The finite element method was used for the numerical solution, in conjunction with the Newton-Raphson method. The numerical examples showed that the model can represent cerebral cortex fold formation and reproduce pathologies related to gyri formation, such as polymicrogyria and lissencephaly.

Shore crabs reveal novel evolutionary attributes of the mushroom body.

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Neural organization of mushroom bodies is largely consistent across insects, whereas the ancestral ground pattern diverges broadly across crustacean lineages resulting in successive loss of columns and the acquisition of domed centers retaining ancestral Hebbian-like networks and aminergic connections. We demonstrate here a major departure from this evolutionary trend in Brachyura, the most recent malacostracan lineage. In the shore crab *Hemigrapsus nudus*, instead of occupying the rostral surface of the lateral protocerebrum, mushroom body calyces are buried deep within it with their columns extending outwards to an expansive system of gyri on the brain's surface. The organization amongst mushroom body neurons reaches extreme elaboration throughout its constituent neuropils. The calyces, columns, and especially the gyri show DC0 immunoreactivity, an indicator of extensive circuits involved in learning and memory.

Brain sulci and gyri: a practical anatomical review.

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Despite technological advances, such as intraoperative MRI, intraoperative sensory and motor monitoring, and awake brain surgery, brain anatomy and its relationship with cranial landmarks still remains the basis of neurosurgery. Our objective is to describe the utility of anatomical knowledge of brain sulci and gyri in neurosurgery. This study was performed on 10 human adult cadaveric heads fixed in formalin and injected with colored silicone rubber. Additionally, using procedures done by the authors

between June 2006 and June 2011, we describe anatomical knowledge of brain sulci and gyri used to manage brain lesions. Knowledge of the brain sulci and gyri can be used (a) to localize the craniotomy procedure, (b) to recognize eloquent areas of the brain, and (c) to identify any given sulcus for access to deep areas of the brain. Despite technological advances, anatomical knowledge of brain sulci and gyri remains essential to perform brain surgery safely and effectively.

The cerebral sulci and gyri.

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The aim of this study was to describe in detail the microanatomy of the cerebral sulci and gyri, clarifying the nomenclature for microneurosurgical purposes. An extensive review of the literature regarding the historical, evolutionary, embryological, and anatomical aspects pertinent to human cerebral sulci and gyri was conducted, with a special focus on microneuroanatomy issues in the field of neurosurgery. An intimate knowledge of the cerebral sulci and gyri is needed to understand neuroimaging studies, as well as to plan and execute current microneurosurgical procedures.

Toward a Common Terminology for the Gyri and Sulci of the Human Cerebral Cortex.

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The gyri and sulci of the human brain were defined by pioneers such as Louis-Pierre Gratiolet and Alexander Ecker, and extensified by, among others, Dejerine (1895) and von Economo and Koskinas (1925). Extensive discussions of the cerebral sulci and their variations were presented by Ono et al. (1990), Duvernoy (1992), Tamraz and Comair (2000), and Rhoton (2007). An anatomical parcellation of the spatially normalized single high resolution T1 volume provided by the Montreal Neurological Institute (MNI; Collins, 1994; Collins et al., 1998) was used for the macroscopical labeling of functional studies (Tzourio-Mazoyer et al., 2002; Rolls et al., 2015). In the standard atlas of the human brain by Mai et al. (2016), the terminology from Mai and Paxinos (2012) is used. It contains an extensively analyzed individual brain hemisphere in the MNI-space. A recent revision of the terminology on the central nervous system in the Terminologia Anatomica (TA, 1998) was made by the Working Group Neuroanatomy of the Federative International Programme for Anatomical Terminology (FIPAT) of the International Federation of Associations of Anatomists (IFAA), and posted online as the Terminologia Neuroanatomica (TNA, 2017: <http://FIPAT.library.dal.ca>) as the official FIPAT terminology. This review deals with the various terminologies for the cerebral gyri and sulci, aiming for a common terminology.